Let’s again identify the components of a data mesh: first the components of the platform and then the data product part.

8.2.2 Identifying the components of the platform

As you learned in chapter 7, a platform consists of a platform kernel, a platform interface, and complements. The first two are the technical components.

In our architecture, the platform team Gray provides four things. One is a data producer Terraform template containing the possible configuration to deploy a new data product into any AWS account. The second is a data consumer Terraform template containing the tools necessary to analyze and transform data from a data consumer perspective. Third, the Gray team provides the central AWS Lake Formation together with an identity and access management (IAM) role repository to let teams grant access to both metadata and data inside their own data products. Finally, the team provides a central PII service, which is a simple API that hashes personalized data that is flagged as such. This is a helper function for data producers, making staying GDPR compliant easy.

Again, the Git repository hosting the templates and its README, as well as the REST API of the PII service, become the platform interface. The templates as well as the inner workings of the PII services and the deployment process for the IAM roles become the platform kernel.

Maintaining this platform might already require one or more full-time engineers.

8.2.3 Identifying the components of the data products

The data consumer Business Analysis team has used the Terraform template to set up a full-fledged stack of tools. The team has Amazon Athena to query the data directly through the AWS interface using SQL. Athena is hooked up to the central Glue catalog and supplied with a role from the central repository. In addition, the Business Analysis team has both SageMaker and Glue for direct usage, deeper exploration of data using notebooks, and possibly transforming data into new data products.